八十八年專門職業及技術人員高等考試試題　　高：14-1　　全一張（正面）

類　　科：冷凍空調工程技師
科　　目：冷凍空調用熱力學與熱傳學
考試時間：二小時　　座號：_____

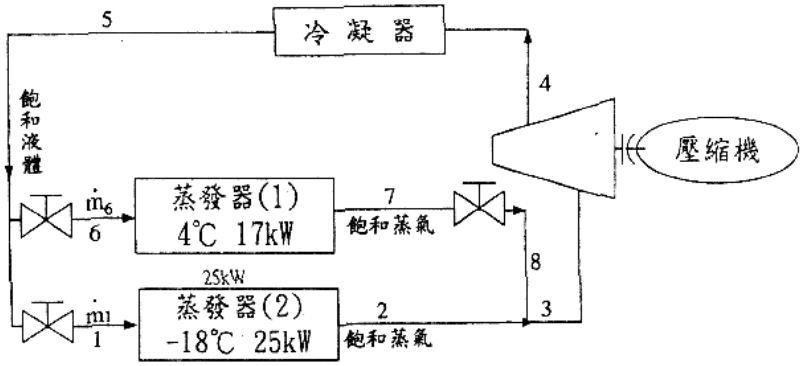
※注意：㈠本試題可以使用電子計算器。
㈡不必抄題，作答時請將試題題號及答案依照順序寫在試卷上，於本試題上作答者，不予計分。

一、有一穩態流動之溼空氣(moist air)，溫度 20°C，壓力 1.05 bars，相對溼度 85%，體積流率 0.3m^3/s。進入一個絕熱(adiabatic)之壓縮機後，其出口溫度 100°C，壓力 2.0 bars，求：
㈠溼空氣之絕對溼度(kg H_2O/kg dry air)及壓縮機出口溼空氣之相對溼度（%）？（10分）
㈡壓縮機所需之功率(KW)？在此總功率中，壓縮乾空氣(dry air)及水蒸汽分別所佔之比率(%)為何？（15分）
註：㈠氣體常數(universal gas constant) R=8.314 kJ/mole K
㈡乾空氣(dry air)分子量=28.97
㈢乾空氣之定壓比熱　C_p=1.004 kJ/kg K
㈣水蒸汽之定壓比熱　C_p=1.872 kJ/kg K

二、㈠何謂理想反向布雷登循環(ideal reversed Brayton cycle)，試繪出其對應之P-V（壓力－比容）及T-S（溫度－熵）圖。（7分）
㈡有一個布雷登空氣冷凍循環，壓縮機及氣渦輪機之等熵效率(isoentropic efficiency)均為 80%，空氣入口壓力為 1 atm，溫度 26°C，壓縮機之壓力比(pressure ratio)為 2，空氣進入氣渦輪機之溫度為 40°C，求本冷凍循環之性能係數(coefficient of performance)？本性能係數比一般的蒸汽壓縮循環(vapor compression cycle)性能係數高或低？為什麼？（18分）
註：㈠空氣之定壓比熱　C_p=1.004 kJ/kg K
㈡空氣之定容比熱　C_v=0.717 kJ/kg K

三、有一磚牆，其熱傳導係數 K_b=1 W/m °C，厚度為 10公分。磚牆之內壁表面溫度為 930°C，外磚牆大氣之溫度為 30°C，大氣之熱對流係數 h=20 W/m^2°C，
㈠求外磚牆表面之溫度（°C）（10分）
㈡若外磚牆表面包覆某種絕熱材料(k=0.1 W/m °C)，欲使絕熱材料之表面溫度不超過 90°C，求所需絕熱材料之最小厚度(cm)？（15分）

四、一個以 R12 為工作流體的理想(ideal)冷凍循環系統，擁有兩個不同操作溫度的蒸發器，如圖所示。其中一個蒸發器的操作溫度為 4°C，提供 17kW 的冷凍能力；另一蒸發器的操作溫度及所提供的冷凍能力分別為 -18°C，25kW。系統中冷凝器的冷凝壓力為 1.0 MPa。求
㈠各個蒸發器的質量流率 $\dot{m}_1$ 及 $\dot{m}_6$ (kg/s)（10分）
㈡本系統的性能係數(coefficient of performance)。（15分）



（請接背面）

八十八年專門職業及技術人員高等考試試題　高：14-1　全一張（背面）

類　科：冷凍空調工程技師

科　目：冷凍空調用熱力學與熱傳學

Properties of Saturated Water (Liquid–Vapor): Temperature Table

Temp. °C	Press. bars	Specific Volume m³/kg: Sat. Liquid $v_f \times 10^3$	Specific Volume m³/kg: Sat. Vapor v_g	Internal Energy kJ/kg: Sat. Liquid u_f	Internal Energy kJ/kg: Sat. Vapor u_g	Enthalpy kJ/kg: Sat. Liquid h_f	Enthalpy kJ/kg: Evap. h_{fg}	Enthalpy kJ/kg: Sat. Vapor h_g	Entropy kJ/kg · K: Sat. Liquid s_f	Entropy kJ/kg · K: Sat. Vapor s_g	Temp. °C
.01	0.00611	1.0002	206.136	0.00	2375.3	0.01	2501.3	2501.4	0.0000	9.1562	.01
4	0.00813	1.0001	157.232	16.77	2380.9	16.78	2491.9	2508.7	0.0610	9.0514	4
5	0.00872	1.0001	147.120	20.97	2382.3	20.98	2489.6	2510.6	0.0761	9.0257	5
6	0.00935	1.0001	137.734	25.19	2383.6	25.20	2487.2	2512.4	0.0912	9.0003	6
8	0.01072	1.0002	120.917	33.59	2386.4	33.60	2482.5	2516.1	0.1212	8.9501	8
10	0.01228	1.0004	106.379	42.00	2388.7	42.00	2477.7	2519.8	0.1510	8.9008	10
11	0.01312	1.0004	99.857	46.20	2390.5	46.20	2475.4	2521.6	0.1658	8.8765	11
12	0.01402	1.0005	93.784	50.41	2391.9	50.41	2473.0	2523.4	0.1806	8.8524	12
13	0.01497	1.0007	88.124	54.60	2393.3	54.60	2470.7	2525.3	0.1953	8.8285	13
14	0.01598	1.0008	82.848	58.79	2394.7	58.80	2468.3	2527.1	0.2099	8.8048	14
15	0.01705	1.0009	77.926	62.99	2396.1	62.99	2465.9	2528.9	0.2245	8.7814	15
16	0.01818	1.0011	73.333	67.18	2397.4	67.19	2463.6	2530.8	0.2390	8.7582	16
17	0.01938	1.0012	69.044	71.38	2398.8	71.38	2461.2	2532.6	0.2535	8.7351	17
18	0.02064	1.0014	65.038	75.57	2400.2	75.58	2458.8	2534.4	0.2679	8.7123	18
19	0.02198	1.0016	61.293	79.76	2401.6	79.77	2456.5	2536.2	0.2823	8.6897	19
20	0.02339	1.0018	57.791	83.95	2402.9	83.96	2454.1	2538.1	0.2966	8.6672	20
21	0.02487	1.0020	54.514	88.14	2404.3	88.14	2451.8	2539.9	0.3109	8.6450	21
22	0.02645	1.0022	51.447	92.32	2405.7	92.33	2449.4	2541.7	0.3251	8.6229	22
23	0.02810	1.0024	48.574	96.51	2407.0	96.52	2447.0	2543.5	0.3393	8.6011	23
24	0.02985	1.0027	45.883	100.70	2408.4	100.70	2444.7	2545.4	0.3534	8.5794	24
25	0.03169	1.0029	43.360	104.88	2409.8	104.89	2442.3	2547.2	0.3674	8.5580	25
26	0.03363	1.0032	40.994	109.06	2411.1	109.07	2439.9	2549.0	0.3814	8.5367	26
27	0.03567	1.0035	38.774	113.25	2412.5	113.25	2437.6	2550.8	0.3954	8.5156	27
28	0.03782	1.0037	36.690	117.42	2413.9	117.43	2435.2	2552.6	0.4093	8.4946	28
29	0.04008	1.0040	34.733	121.60	2415.2	121.61	2432.8	2554.5	0.4231	8.4739	29
30	0.04246	1.0043	32.894	125.78	2416.6	125.79	2430.5	2556.3	0.4369	8.4533	30
31	0.04496	1.0046	31.165	129.96	2418.0	129.97	2428.1	2558.1	0.4507	8.4329	31
32	0.04759	1.0050	29.540	134.14	2419.3	134.15	2425.7	2559.9	0.4644	8.4127	32
33	0.05034	1.0053	28.011	138.32	2420.7	138.33	2423.4	2561.7	0.4781	8.3927	33
34	0.05324	1.0056	26.571	142.50	2422.0	142.50	2421.0	2563.5	0.4917	8.3728	34
35	0.05628	1.0060	25.216	146.67	2423.4	146.68	2418.6	2565.3	0.5053	8.3531	35
36	0.05947	1.0063	23.940	150.85	2424.7	150.86	2416.2	2567.1	0.5188	8.3336	36
38	0.06632	1.0071	21.602	159.20	2427.4	159.21	2411.5	2570.7	0.5458	8.2950	38
40	0.07384	1.0078	19.523	167.56	2430.1	167.57	2406.7	2574.3	0.5725	8.2570	40
45	0.09593	1.0099	15.258	188.44	2436.8	188.45	2394.8	2583.2	0.6387	8.1648	45

TABLE A-2 (*Continued*)

Temp. °C	Press. bars	Specific Volume m³/kg: Sat. Liquid $v_f \times 10^3$	Specific Volume m³/kg: Sat. Vapor v_g	Internal Energy kJ/kg: Sat. Liquid u_f	Internal Energy kJ/kg: Sat. Vapor u_g	Enthalpy kJ/kg: Sat. Liquid h_f	Enthalpy kJ/kg: Evap. h_{fg}	Enthalpy kJ/kg: Sat. Vapor h_g	Entropy kJ/kg · K: Sat. Liquid s_f	Entropy kJ/kg · K: Sat. Vapor s_g	Temp. °C
50	.1235	1.0121	12.032	209.32	2443.5	209.33	2382.7	2592.1	.7038	8.0763	50
55	.1576	1.0146	9.568	230.21	2450.1	230.23	2370.7	2600.9	.7679	7.9913	55
60	.1994	1.0172	7.671	251.11	2456.6	251.13	2358.5	2609.6	.8312	7.9096	60
65	.2503	1.0199	6.197	272.02	2463.1	272.06	2346.2	2618.3	.8935	7.8310	65
70	.3119	1.0228	5.042	292.95	2469.6	292.98	2333.8	2626.8	.9549	7.7553	70
75	.3858	1.0259	4.131	313.90	2475.9	313.93	2321.4	2635.3	1.0155	7.6824	75
80	.4739	1.0291	3.407	334.86	2482.2	334.91	2308.8	2643.7	1.0753	7.6122	80
85	.5783	1.0325	2.828	355.84	2488.4	355.90	2296.0	2651.9	1.1343	7.5445	85
90	.7014	1.0360	2.361	376.85	2494.5	376.92	2283.2	2660.1	1.1925	7.4791	90
95	.8455	1.0397	1.982	397.88	2500.6	397.96	2270.2	2668.1	1.2500	7.4159	95
100	1.014	1.0435	1.6729	418.94	2506.5	419.04	2257.0	2676.1	1.3069	7.3549	100
110	1.433	1.0516	1.210	461.14	2518.1	461.30	2230.2	2691.5	1.4185	7.2387	110
120	1.985	1.0603	0.8919	503.50	2529.3	503.71	2202.6	2706.3	1.5276	7.1296	120
130	2.701	1.0697	0.6685	546.02	2539.9	546.31	2174.2	2720.5	1.6344	7.0269	130
140	3.613	1.0797	0.5089	588.74	2550.0	589.13	2144.7	2733.9	1.7391	6.9299	140
150	4.758	1.0905	0.3928	631.68	2559.5	632.20	2114.3	2746.5	1.8418	6.8379	150
160	6.178	1.1020	0.3071	674.86	2568.4	675.55	2082.6	2758.1	1.9427	6.7502	160
170	7.917	1.1143	0.2428	718.33	2576.5	719.21	2049.5	2768.7	2.0419	6.6663	170
180	10.02	1.1274	0.1941	762.09	2583.7	763.22	2015.0	2778.2	2.1396	6.5857	180
190	12.54	1.1414	0.1565	806.19	2590.0	807.62	1978.8	2786.4	2.2359	6.5079	190
200	15.54	1.1565	0.1274	850.65	2595.3	852.45	1940.7	2793.2	2.3309	6.4323	200
210	19.06	1.1726	0.1044	895.53	2599.5	897.76	1900.7	2798.5	2.4248	6.3585	210
220	23.18	1.1900	0.08619	940.87	2602.4	943.62	1858.5	2802.1	2.5178	6.2861	220
230	27.95	1.2088	0.07158	986.74	2603.9	990.12	1813.8	2804.0	2.6099	6.2146	230
240	33.44	1.2291	0.05976	1033.2	2604.0	1037.3	1766.5	2803.8	2.7015	6.1437	240
250	39.73	1.2512	0.05013	1080.4	2602.4	1085.4	1716.2	2801.5	2.7927	6.0730	250
260	46.88	1.2755	0.04221	1128.4	2599.0	1134.4	1662.5	2796.6	2.8838	6.0019	260
270	54.99	1.3023	0.03564	1177.4	2593.7	1184.5	1605.2	2789.7	2.9751	5.9301	270
280	64.12	1.3321	0.03017	1227.5	2586.1	1236.0	1543.6	2779.6	3.0668	5.8571	280
290	74.36	1.3656	0.02557	1278.9	2576.0	1289.1	1477.1	2766.2	3.1594	5.7821	290
300	85.81	1.4036	0.02167	1332.0	2563.0	1344.0	1404.9	2749.0	3.2534	5.7045	300
320	112.7	1.4988	0.01549	1444.6	2525.5	1461.5	1238.6	2700.1	3.4480	5.5362	320
340	145.9	1.6379	0.01080	1570.3	2464.6	1594.2	1027.9	2622.0	3.6594	5.3357	340
360	186.5	1.8925	0.006945	1725.2	2351.5	1760.5	720.5	2481.0	3.9147	5.0526	360
374.14	220.9	3.155	0.003155	2029.6	2029.6	2099.3	0	2099.3	4.4298	4.4298	374.14

Source: Tables A-2 through A-6 were adapted from K. Wark, *Thermodynamics*, 4th ed., McGraw-Hill, New York, 1983, as extracted from J. H. Keenan, F. G. Keyes, P. G. Hill, and J. G. Moore, *Steam Tables*, Wiley, New York, 1969.

TABLE A-3 Properties of Saturated Refrigerant 12 (Liquid–Vapor): Temperature Table

Temp. °C	Press. bars	Specific Volume m³/kg: Sat. Liquid $v_f \times 10^3$	Specific Volume m³/kg: Sat. Vapor v_g	Internal Energy kJ/kg: Sat. Liquid u_f	Internal Energy kJ/kg: Sat. Vapor u_g	Enthalpy kJ/kg: Sat. Liquid h_f	Enthalpy kJ/kg: Evap. h_{fg}	Enthalpy kJ/kg: Sat. Vapor h_g	Entropy kJ/kg · K: Sat. Liquid s_f	Entropy kJ/kg · K: Sat. Vapor s_g	Temp. °C
−40	0.6417	0.6595	0.24191	−0.04	154.07	0.00	169.59	169.59	0.0000	0.7274	−40
−35	0.8071	0.6656	0.19540	4.37	156.13	4.42	167.48	171.90	0.0187	0.7219	−35
−30	1.0041	0.6720	0.15938	8.79	158.20	8.86	165.33	174.20	0.0371	0.7170	−30
−28	1.0927	0.6746	0.14728	10.58	159.02	10.65	164.46	175.11	0.0444	0.7153	−28
−26	1.1872	0.6773	0.13628	12.35	159.84	12.43	163.59	176.02	0.0517	0.7135	−26
−25	1.2368	0.6786	0.13117	13.25	160.26	13.33	163.15	176.48	0.0552	0.7126	−25
−24	1.2880	0.6800	0.12628	14.13	160.67	14.22	162.71	176.93	0.0589	0.7119	−24
−22	1.3953	0.6827	0.11717	15.92	161.48	16.02	161.82	177.83	0.0660	0.7103	−22
−20	1.5093	0.6855	0.10885	17.72	162.31	17.82	160.92	178.74	0.0731	0.7087	−20
−18	1.6304	0.6883	0.10124	19.51	163.12	19.62	160.01	179.63	0.0802	0.7073	−18
−15	1.8260	0.6926	0.09102	22.20	164.35	22.33	158.64	180.97	0.0906	0.7051	−15
−10	2.1912	0.7000	0.07665	26.72	166.39	26.87	156.31	183.19	0.1080	0.7019	−10
−5	2.6096	0.7078	0.06496	31.27	168.42	31.45	153.93	185.37	0.1251	0.6991	−5
0	3.0861	0.7159	0.05539	35.83	170.44	36.05	151.48	187.53	0.1420	0.6965	0
4	3.5124	0.7227	0.04895	39.51	172.04	39.76	149.47	189.23	0.1553	0.6946	4
8	3.9815	0.7297	0.04340	43.21	173.63	43.50	147.41	190.91	0.1686	0.6929	8
12	4.4962	0.7370	0.03860	46.93	175.20	47.26	145.30	192.56	0.1817	0.6913	12
16	5.0591	0.7446	0.03442	50.67	176.78	51.05	143.14	194.19	0.1948	0.6898	16
20	5.6729	0.7525	0.03078	54.44	178.32	54.87	140.91	195.78	0.2078	0.6884	20
24	6.3405	0.7607	0.02759	58.25	179.85	58.73	138.61	197.34	0.2207	0.6871	24
26	6.6954	0.7650	0.02614	60.17	180.61	60.68	137.44	198.11	0.2271	0.6865	26
28	7.0648	0.7694	0.02478	62.09	181.36	62.63	136.24	198.87	0.2335	0.6859	28
30	7.4490	0.7739	0.02351	64.01	182.11	64.59	135.03	199.62	0.2400	0.6853	30
32	7.8485	0.7785	0.02231	65.96	182.85	66.57	133.79	200.36	0.2463	0.6847	32
34	8.2636	0.7832	0.02118	67.90	183.59	68.55	132.53	201.09	0.2527	0.6842	34
36	8.6948	0.7880	0.02012	69.86	184.31	70.55	131.25	201.80	0.2591	0.6836	36
38	9.1423	0.7929	0.01912	71.84	185.03	72.56	129.94	202.51	0.2655	0.6831	38
40	9.6065	0.7980	0.01817	73.82	185.74	74.59	128.61	203.20	0.2718	0.6825	40
42	10.088	0.8033	0.01728	75.82	186.45	76.63	127.25	203.88	0.2782	0.6820	42
44	10.587	0.8086	0.01644	77.82	187.13	78.68	125.87	204.54	0.2845	0.6814	44
48	11.629	0.8199	0.01488	81.88	188.51	82.83	123.00	205.83	0.2973	0.6802	48
52	12.766	0.8318	0.01349	86.00	189.83	87.06	119.99	207.05	0.3101	0.6791	52
56	13.972	0.8445	0.01224	90.18	191.10	91.36	116.84	208.20	0.3229	0.6779	56
60	15.259	0.8581	0.01111	94.43	192.31	95.74	113.52	209.26	0.3358	0.6765	60
112	41.155	1.792	0.00179	175.98	175.98	183.35	0	183.35	0.5687	0.5687	112

Source: Tables A-7 through A-9 are adapted from K. Wark, *Thermodynamics*, 4th ed., McGraw-Hill, New York, 1983, as based on data from Freon Products Division, E. I. du Pont de Nemours & Company, 1969.

Properties of Superheated Refrigerant 12 Vapor

T °C	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg · K
	p = 0.6 bar = 0.06 MPa (T_{sat} = −43.42°C)				p = 1.0 bar = 0.10 MPa (T_{sat} = −30.10°C)			
Sat.	0.2575	153.69	168.94	0.7290	0.1600	158.15	174.15	0.7171
−40	0.2593	154.16	169.72	0.7324				
−20	0.2838	163.91	180.94	0.7785	0.1677	163.22	179.99	0.7406
0	0.3079	174.05	192.52	0.8225	0.1827	173.50	191.77	0.7854
10	0.3198	179.26	198.45	0.8439	0.1900	178.77	197.77	0.8070
20	0.3317	184.57	204.47	0.8647	0.1973	184.12	203.85	0.8281
30	0.3435	189.96	210.57	0.8852	0.2045	189.57	210.02	0.8488
40	0.3552	195.46	216.77	0.9053	0.2117	195.09	216.26	0.8691
50	0.3670	201.02	223.04	0.9251	0.2188	200.70	222.58	0.8889
60	0.3787	206.69	229.41	0.9444	0.2260	206.38	228.98	0.9084
80	0.4020	218.25	242.37	0.9822	0.2401	218.00	242.01	0.9464
	p = 1.4 bars = 0.14 MPa (T_{sat} = −21.91°C)				p = 1.8 bars = 0.18 MPa (T_{sat} = −15.38°C)			
Sat.	0.1168	161.52	177.87	0.7102	0.0922	164.20	180.80	0.7054
−20	0.1179	162.50	179.01	0.7147				
−10	0.1235	167.69	184.97	0.7378	0.0925	164.39	181.03	0.7181
0	0.1289	172.94	190.99	0.7602	0.0991	172.37	190.21	0.7408
10	0.1343	178.28	197.08	0.7821	0.1034	177.77	196.38	0.7630
20	0.1397	183.67	203.23	0.8035	0.1076	183.23	202.60	0.7846
30	0.1449	189.17	209.46	0.8243	0.1118	188.77	208.89	0.8057
40	0.1502	194.72	215.75	0.8447	0.1160	194.35	215.23	0.8263
50	0.1553	200.38	222.12	0.8648	0.1201	200.02	221.64	0.8464
60	0.1605	206.08	228.55	0.8844	0.1241	205.78	228.12	0.8662
80	0.1707	217.74	241.64	0.9225	0.1322	217.47	241.27	0.9045
100	0.1809	229.67	255.00	0.9593	0.1402	229.45	254.69	0.9414

TABLE A-9 (*Continued*)

T °C	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg · K	v m³/kg	u kJ/kg	h kJ/kg	s kJ/kg · K
	p = 8.0 bars = 0.80 MPa (T_{sat} = 32.76°C)				p = 9.0 bars = 0.90 MPa (T_{sat} = 37.37°C)			
Sat.	0.02188	183.13	200.63	0.6845	0.01942	184.81	202.29	0.6832
40	0.02283	187.81	206.37	0.7021	0.01974	186.55	204.32	0.6897
50	0.02407	194.19	213.45	0.7233	0.02081	193.10	211.92	0.7136
60	0.02525	200.52	220.72	0.7474	0.02201	199.56	219.37	0.7363
80	0.02748	213.13	235.11	0.7894	0.02407	212.37	234.03	0.7790
100	0.02959	225.77	249.44	0.8289	0.02601	225.13	248.54	0.8190
120	0.03162	238.51	263.81	0.8664	0.02785	237.97	263.03	0.8569
140	0.03359	251.39	278.26	0.9022	0.02964	250.90	277.58	0.8930
160	0.03552	264.41	292.83	0.9367	0.03138	263.99	292.23	0.9276
180	0.03742	277.60	307.54	0.9699	0.03309	277.23	307.01	0.9609
	p = 10.0 bars = 1.00 MPa (T_{sat} = 41.64°C)				p = 12.0 bars = 1.20 MPa (T_{sat} = 49.31°C)			
Sat.	0.01744	186.32	203.76	0.6820	0.01441	188.95	206.24	0.6799
50	0.01837	191.95	210.32	0.7026	0.01448	189.43	206.81	0.6816
60	0.01941	198.56	217.97	0.7259	0.01546	196.41	214.96	0.7065
80	0.02134	211.57	232.91	0.7695	0.01722	209.91	230.57	0.7520
100	0.02313	224.48	247.61	0.8100	0.01881	223.13	245.70	0.7937
120	0.02484	237.41	262.25	0.8482	0.02030	236.27	260.63	0.8326
140	0.02647	250.43	276.90	0.8845	0.02172	249.45	275.51	0.8696
160	0.02807	263.56	291.63	0.9193	0.02309	263.70	290.41	0.9048
180	0.02963	276.84	306.47	0.9528	0.02443	276.05	305.37	0.9385
200	0.03116	290.26	321.42	0.9851	0.02574	289.55	320.44	0.9711